Let us redefine, for convenience, the *non-negative homogeneity* property (Neyshabur et al., 2015; Lafond et al., 2016) below. Note that beside this property, the reason for studying the rectified linear activation is for its widespread adoption (Krizhevsky et al., 2012; Simonyan & Zisserman, 2015; Szegedy et al., 2015; He et al., 2016).

Definition 4. *A given a function ϕ is* non-negative homogeneous *if*

$$\forall (z, \alpha) \in \mathbb{R} \times \mathbb{R}^+, \phi(\alpha z) = \alpha \phi(z)$$

.

Theorem 1. *The rectified function $\phi_{rect}(x) = \max(x, 0)$ is non-negative homogeneous.*

Proof. Follows trivially from the constraint that $\alpha > 0$, given that $x > 0 \Rightarrow \alpha x > 0$, iff $\alpha > 0$.

□

For a deep rectified neural network it means that:

$$\phi_{rect}\big(x \cdot (\alpha\theta_1)\big) \cdot \theta_2 = \phi_{rect}(x \cdot \theta_1) \cdot (\alpha\theta_2),$$

meaning that for this one (hidden) layer neural network, the parameters $(\alpha\theta_1, \theta_2)$ is observationally equivalent to $(\theta_1, \alpha\theta_2)$. This observational equivalence similarly holds for convolutional layers.

Given this non-negative homogeneity, if $(\theta_1, \theta_2) \neq (0, 0)$ then $\big\{(\alpha\theta_1, \alpha^{-1}\theta_2), \alpha > 0\big\}$ is an infinite set of observationally equivalent parameters, inducing a strong non-identifiability in this learning scenario. Other models like *deep linear networks* (Saxe et al., 2013), *leaky rectifiers* (He et al., 2015) or *maxout networks* (Goodfellow et al., 2013) also have this non-negative homogeneity property.

In what follows we will rely on such transformations, in particular we will rely on the following definition:

Definition 5. *For a single hidden layer rectifier feedforward network we define the family of transformations*

$$T_\alpha : (\theta_1, \theta_2) \mapsto (\alpha\theta_1, \alpha^{-1}\theta_2)$$

which we refer to as a α-scale transformation.

Note that a α-scale transformation will not affect the generalization, as the behavior of the function is identical. Also while the transformation is only defined for a single layer rectified feedforward network, it can trivially be extended to any architecture having a single rectified network as a submodule, e.g. a deep rectified feedforward network. For simplicity and readability we will rely on this definition.

4 Deep Rectified networks and flat minima

In this section we exploit the resulting strong non-identifiability to showcase a few shortcomings of some definitions of flatness. Although α-scale transformation does not affect the function represented, it allows us to significantly decrease several measures of flatness. For another definition of flatness, α-scale transformation show that all minima are equally flat.

4.1 Volume ϵ-flatness

Theorem 2. *For a one-hidden layer rectified neural network of the form*

$$y = \phi_{rect}(x \cdot \theta_1) \cdot \theta_2,$$

and a minimum $\theta = (\theta_1, \theta_2)$, such that $\theta_1 \neq 0$ and $\theta_2 \neq 0$, $\forall \epsilon > 0$ $C(L, \theta, \epsilon)$ has an infinite volume.

We will not consider the solution θ where any of the weight matrices θ_1, θ_2 is zero, $\theta_1 = 0$ or $\theta_2 = 0$, as it results in a constant function which we will assume to give poor training performance. For $\alpha > 0$, the α-scale transformation $T_\alpha : (\theta_1, \theta_2) \mapsto (\alpha\theta_1, \alpha^{-1}\theta_2)$ has Jacobian determinant $\alpha^{n_1 - n_2}$, where once again $n_1 = dim\big(\text{vec}(\theta_1)\big)$ and $n_2 = dim\big(\text{vec}(\theta_2)\big)$. Note that the Jacobian determinant of this linear transformation is the change in the volume induced by T_α and $T_\alpha \circ T_\beta = T_{\alpha\beta}$. We show below that there is a connected region containing θ with infinite volume and where the error remains approximately constant.

Proof. We will first introduce a small region with approximately constant error around θ with non-zero volume. Given $\epsilon > 0$ and if we consider the loss function continuous with respect to the parameter, $C(L, \theta, \epsilon)$ is an open set containing θ. Since we also have $\theta_1 \neq 0$ and $\theta_2 \neq 0$, let $r > 0$ such that the $\mathcal{L}_\infty$ ball $B_\infty(r, \theta)$ is in $C(L, \theta, \epsilon)$ and has empty intersection with $\{\theta', \theta'_1 = 0\}$. Let $v = (2r)^{n_1 + n_2} > 0$ the volume of $B_\infty(r, \theta)$.

Since the Jacobian determinant of T_α is the multiplicative change of induced by T_α, the volume of $T_\alpha\big(B_\infty(r, \theta)\big)$ is $v\alpha^{n_1 - n_2}$. If $n_1 \neq n_2$, we can arbitrarily grow the volume of $T_\alpha\big(B_\infty(r, \theta)\big)$, with error within an ϵ-interval of $L(\theta)$, by having α tends to $+\infty$ if $n_1 > n_2$ or to 0 otherwise.

If $n_1 = n_2$, $\forall \alpha' > 0, T_{\alpha'}\Big(B_\infty(r, \theta)\Big)$ has volume v. Let $C' = \bigcup_{\alpha' > 0} T_{\alpha'}\Big(B_\infty(r, \theta)\Big)$. C' is a connected region where the error remains approximately constant, i.e. within an ϵ-interval of $L(\theta)$.

Let $\alpha = 2\frac{\|\theta_1\|_\infty + r}{\|\theta_1\|_\infty - r}$. Since

$$B_\infty(r, \theta) = B_\infty(r, \theta_1) \times B_\infty(r, \theta_2),$$

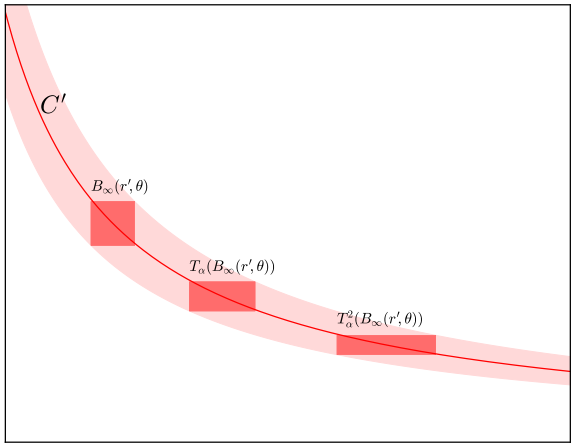

Figure 3: An illustration of how we build different disjoint volumes using T_α. In this two-dimensional example, $T_\alpha\big(B_\infty(r',\theta)\big)$ and $B_\infty(r',\theta)$ have the same volume. $B_\infty(r',\theta), T_\alpha\big(B_\infty(r',\theta)\big), T_\alpha^2\big(B_\infty(r',\theta)\big), \dots$ will therefore be a sequence of disjoint constant volumes. C' will therefore have an infinite volume. Best seen with colors.

where $\times$ is the Cartesian set product, we have

$$T_\alpha\big(B_\infty(r,\theta)\big) = B_\infty(\alpha r, \alpha\theta_1) \times B_\infty(\alpha^{-1} r, \alpha^{-1}\theta_2).$$

Therefore, $T_\alpha\big(B_\infty(r,\theta)\big) \cap B_\infty(r,\theta) = \emptyset$ (see Figure 3).

Similarly, $B_\infty(r,\theta), T_\alpha\big(B_\infty(r,\theta)\big), T_\alpha^2\big(B_\infty(r,\theta)\big), \dots$ are disjoint and have volume v. We have also $T_\alpha^k\big(B_\infty(r',\theta)\big) = T_{\alpha^k}\big(B_\infty(r',\theta)\big) \in C'$. The volume of C' is then lower bounded by $0 < v + v + v + \cdots$ and is therefore infinite. $C(L,\theta,\epsilon)$ has then infinite volume too, making the volume ϵ-flatness of θ infinite. □

This theorem can generalize to rectified neural networks in general with a similar proof. Given that every minimum has an infinitely large region (volume-wise) in which the error remains approximately constant, that means that every minimum would be infinitely flat according to the volume ϵ-flatness. Since all minima are equally flat, it is not possible to use volume ϵ-flatness to gauge the generalization property of a minimum.

4.2 Hessian-based measures

The non-Euclidean geometry of the parameter space, coupled with the manifolds of observationally equal behavior of the model, allows one to move from one region of the parameter space to another, changing the curvature of the model without actually changing the function. This approach has been used with success to improve optimization, by moving from a region of high curvature to a region of well behaved curvature (e.g. Desjardins et al., 2015; Salimans & Kingma, 2016). In this section we look at two widely used measures of the Hessian, the spectral radius and trace, showing that either of these values can be manipulated without actually changing the behavior of the function. If the flatness of a minimum is defined by any of these quantities, then it could also be easily manipulated.

Theorem 3. *The gradient and Hessian of the loss L with respect to θ can be modified by T_α.*

Proof.

$$L(\theta_1,\theta_2) = L(\alpha\theta_1, \alpha^{-1}\theta_2),$$

we have then by differentiation

$$(\nabla L)(\theta_1,\theta_2) = (\nabla L)(\alpha\theta_1,\alpha^{-1}\theta_2)\begin{bmatrix} \alpha\mathbb{I}_{n_1} & 0 \\ 0 & \alpha^{-1}\mathbb{I}_{n_2}\end{bmatrix}$$
$$\Leftrightarrow (\nabla L)(\alpha\theta_1,\alpha^{-1}\theta_2) = (\nabla L)(\theta_1,\theta_2)\begin{bmatrix} \alpha^{-1}\mathbb{I}_{n_1} & 0 \\ 0 & \alpha\mathbb{I}_{n_2}\end{bmatrix}$$

and

$$(\nabla^2 L)(\alpha\theta_1,\alpha^{-1}\theta_2) = \begin{bmatrix} \alpha^{-1}\mathbb{I}_{n_1} & 0 \\ 0 & \alpha\mathbb{I}_{n_2}\end{bmatrix}(\nabla^2 L)(\theta_1,\theta_2)\begin{bmatrix} \alpha^{-1}\mathbb{I}_{n_1} & 0 \\ 0 & \alpha\mathbb{I}_{n_2}\end{bmatrix}.$$

□

Sharpest direction Through these transformations we can easily find, for any critical point which is a minimum with non-zero Hessian, an observationally equivalent parameter whose Hessian has an arbitrarily large spectral norm.

Theorem 4. *For a one-hidden layer rectified neural network of the form*

$$y = \phi_{rect}(x\cdot\theta_1)\cdot\theta_2,$$

and critical point $\theta = (\theta_1,\theta_2)$ being a minimum for L, such that $(\nabla^2 L)(\theta) \neq 0$, $\forall M > 0, \exists\alpha > 0, \left|\left|\left|(\nabla^2 L)\big(T_\alpha(\theta)\big)\right|\right|\right|_2 \geq M$ where $\left|\left|\left|(\nabla^2 L)\big(T_\alpha(\theta)\big)\right|\right|\right|_2$ is the spectral norm of $(\nabla^2 L)\big(T_\alpha(\theta)\big)$.

Proof. The trace of a symmetric matrix is the sum of its eigenvalues and a real symmetric matrix can be diagonalized in $\mathbb{R}$, therefore if the Hessian is non-zero, there is one non-zero positive diagonal element. Without loss of generality, we will assume that this non-zero element of value $\gamma > 0$ corresponds to an element in θ_1. Therefore the Frobenius norm $\left|\left|\left|(\nabla^2 L)\big(T_\alpha(\theta)\big)\right|\right|\right|_F$ of

$$(\nabla^2 L)(\alpha\theta_1,\alpha^{-1}\theta_2) = \begin{bmatrix} \alpha^{-1}\mathbb{I}_{n_1} & 0 \\ 0 & \alpha\mathbb{I}_{n_2}\end{bmatrix}(\nabla^2 L)(\theta_1,\theta_2)\begin{bmatrix} \alpha^{-1}\mathbb{I}_{n_1} & 0 \\ 0 & \alpha\mathbb{I}_{n_2}\end{bmatrix}.$$

is lower bounded by $\alpha^{-2}\gamma$.

Since all norms are equivalent in finite dimension, there exists a constant $r > 0$ such that $r|||A|||_F \leq |||A|||_2$ for all symmetric matrices A. So by picking $\alpha < \sqrt{\frac{r\gamma}{M}}$, we are guaranteed that $\left|\left|\left|(\nabla^2 L)\big(T_\alpha(\theta)\big)\right|\right|\right|_2 \geq M$. □

Any minimum with non-zero Hessian will be observationally equivalent to a minimum whose Hessian has an arbitrarily large spectral norm. Therefore for any minimum in the loss function, if there exists another minimum that generalizes better then there exists another minimum that generalizes better and is also sharper according the spectral norm of the Hessian. The spectral norm of critical points' Hessian becomes as a result less relevant as a measure of potential generalization error. Moreover, since the spectral norm lower bounds the trace for a positive semi-definite symmetric matrix, the same conclusion can be drawn for the trace.

Many directions However, some notion of sharpness might take into account the entire eigenspectrum of the Hessian as opposed to its largest eigenvalue, for instance, Chaudhari et al. (2017) describe the notion of *wide valleys*, allowing the presence of very few large eigenvalues. We can generalize the transformations between observationally equivalent parameters to deeper neural networks with $K-1$ hidden layers: for $\alpha_k > 0, T_\alpha : (\theta_k)_{k\leq K} \mapsto (\alpha_k\theta_k)_{k\in K}$ with $\prod_{k=1}^K \alpha_k = 1$. If we define

$$D_\alpha = \begin{bmatrix} \alpha_1^{-1}\mathbb{I}_{n_1} & 0 & \cdots & 0 \\ 0 & \alpha_2^{-1}\mathbb{I}_{n_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_K^{-1}\mathbb{I}_{n_K} \end{bmatrix}$$

then the first and second derivatives at $T_\alpha(\theta)$ will be

$$\begin{aligned}(\nabla L)\big(T_\alpha(\theta)\big) =& (\nabla L)(\theta)D_\alpha \\ (\nabla^2 L)\big(T_\alpha(\theta)\big) =& D_\alpha(\nabla^2 L)(\theta)D_\alpha.\end{aligned}$$

We will show to which extent you can increase several eigenvalues of $(\nabla^2 L)\big(T_\alpha(\theta)\big)$ by varying α.

Definition 6. *For each $n \times n$ matrix A, we define the vector $\lambda(A)$ of sorted singular values of A with their multiplicity $\lambda_1(A) \geq \lambda_2(A) \geq \cdots \geq \lambda_n(A)$.*

If A is symmetric positive semi-definite, $\lambda(A)$ is also the vector of its sorted eigenvalues.

Theorem 5. *For a $(K-1)$-hidden layer rectified neural network of the form*

$$y = \phi_{rect}(\phi_{rect}(\cdots\phi_{rect}(x\cdot\theta_1)\cdots)\cdot\theta_{K-1})\cdot\theta_K,$$

and critical point $\theta = (\theta_k)_{k\leq K}$ being a minimum for L, such that $(\nabla^2 L)(\theta)$ has rank $r = rank\big((\nabla^2 L)(\theta)\big)$, $\forall M > 0, \exists\alpha > 0$ such that $\Big(r - \min_{k\leq K}(n_k)\Big)$ eigenvalues are greater than M.

Proof. For simplicity, we will note $\sqrt{M}$ the *principal square root* of a symmetric positive-semidefinite matrix M. The eigenvalues of $\sqrt{M}$ are the square root of the eigenvalues of M and are its *singular values*. By definition, the *singular values* of $\sqrt{(\nabla^2 L)(\theta)}D_\alpha$ are the square root of the eigenvalues of $D_\alpha(\nabla^2 L)(\theta)D_\alpha$. Without loss of generality, we consider $\min_{k\leq K}(n_k) = n_K$ and choose $\forall k < K, \alpha_k = \beta^{-1}$ and $\alpha_K = \beta^{K-1}$. Since D_α and $\sqrt{(\nabla^2 L)(\theta)}$ are positive symmetric semi-definite matrices, we can apply the multiplicative Horn inequalities (Klyachko, 2000) on singular values of the product $\sqrt{(\nabla^2 L)(\theta)}D_\alpha$:

$$\begin{aligned}&\forall i \leq n, j \leq (n - n_K), \\ &\qquad \lambda_{i+j-n}\big((\nabla^2 L)(\theta)D_\alpha^2\big) \geq \lambda_i\big((\nabla^2 L)(\theta)\big)\beta^2.\end{aligned}$$

By choosing $\beta > \sqrt{\frac{M}{\lambda_r\big((\nabla^2 L)(\theta)\big)}}$, since we have $\forall i \leq r, \lambda_i\big((\nabla^2 L)(\theta)\big) \geq \lambda_r\big((\nabla^2 L)(\theta)\big) > 0$ we can conclude that

$$\begin{aligned}&\forall i \leq (r - n_K), \\ &\lambda_i\big((\nabla^2 L)(\theta)D_\alpha^2\big) \geq \lambda_{i+n_k}\big((\nabla^2 L)(\theta)\big)\beta^2 \\ &\qquad\qquad \geq \lambda_r\big((\nabla^2 L)(\theta)\big)\beta^2 > M.\end{aligned}$$

□

It means that there exists an observationally equivalent parameter with at least $\Big(r - \min_{k\leq K}(n_k)\Big)$ arbitrarily large eigenvalues. Since Sagun et al. (2016) seems to suggests that rank deficiency in the Hessian is due to over-parametrization of the model, one could conjecture that $\Big(r - \min_{k\leq K}(n_k)\Big)$ can be high for thin and deep neural networks, resulting in a majority of large eigenvalues. Therefore, it would still be possible to obtain an equivalent parameter with large Hessian eigenvalues, i.e. sharp in multiple directions.

4.3 ϵ-sharpness

We have redefined for $\epsilon > 0$ the ϵ-sharpness of Keskar et al. (2017) as follow

$$\frac{\max_{\theta'\in B_2(\epsilon,\theta)}\big(L(\theta') - L(\theta)\big)}{1 + L(\theta)}$$

where $B_2(\epsilon, \theta)$ is the Euclidean ball of radius ϵ centered on θ. This modification will demonstrate more clearly the issues of that metric as a measure of probable generalization. If we use $K = 2$ and (θ_1, θ_2) corresponding to a non-constant function, i.e. $\theta_1 \neq 0$ and $\theta_2 \neq 0$, then we can